

Brain Inspired Computing

Moving from Known to the Unknown Worlds

Over the last few decades technology has integrated itself into our daily lives in ways which make us totally reliant on it for our day-to-day tasks. Undoubtedly, technology has only gotten smarter. Considering the trajectory it currently follows, it is only a matter of time until it is able to mimic the way we think and perform our daily tasks. Continue reading to know more.

We are living in an extraordinary time when we are trying to recreate a human brain in machine. Human brain is the most sophisticated organ that nature has given us. Building something which can equal that is a very difficult feat to achieve.

Over the last few years, a lot of studies and findings have been done but that does not seem to be enough by any means. Our limited understanding of this complex organ makes it even harder to mimic all of its functions precisely. Yet, we are determined to make something that can at least be close to the real thing.

Technology landscape

Figure 1 shows an analogue from

the perspective of the technology landscape. It shows that a personal computer and a wired internet connection can be used to connect to the external world.

The horizontal dimension in the figure shows that with the help of larger systems we are able to integrate data altogether on a large scale storage and process that data through cloud. The vertical direction contains the devices and technologies that provide a plethora of functionalities.

For example, our cell phones can sense, display, communicate via wireless systems and the internet while providing a great computational power for data processing applications. The computing domain can change from a static and virtual environment

to the real world which is highly dynamic in nature. By combining these two dimensions we envision intelligent systems.

Intelligence and artificial intelligence

The concept of intelligence has been discussed by people for many years, even as far back as the 18th century. According to Wilhelm Wundt, "Our mind is so fortunately equipped that it brings us the most important bases for our thought with our having the least knowledge of this work of elaboration." Clearly, we are curious about intelligence, but we do not know much about our brains and where does it actually come from.

People have become more curious

about the intelligence itself through the years of exploration, which has led them to build intelligent systems we today call artificial intelligence (AI). The history of AI can be traced back to a summer workshop back in 1956. It is widely considered as the founding event of AI as a field.

According to Wikipedia, "It is a theory and development of computer systems able

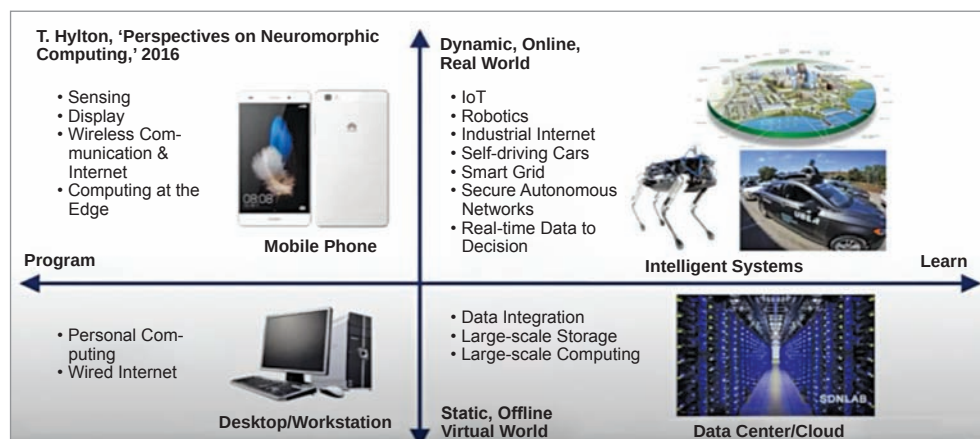


Figure 1: Technology landscape